

Python Lab: Working with Text Files (Students & Grades)

Lab Objective

In this lab, you will practice working with **text files (.txt)** in Python.
You will learn how to:

- Create text files using Python
- Write structured data to files
- Read and process data from files
- Link data between multiple files using an ID

This lab simulates a **real-world scenario** of managing student data.

Lab Files

You will work with **two text files**:

1 students.txt

This file contains basic student information.

Format:

```
student_id,name
```

Example:

```
1,Ahmed Ali  
2,Sara Mohamed  
3,Omar Hassan
```

2 grades.txt

This file contains students' grades for different subjects.

Format:

```
student_id,subject,grade
```

Example:

```
1,Python,85
1,Math,90
2,Python,78
```

Lab Tasks

Task 1

Create a Python script that generates the following files:

- `students.txt`
- `grades.txt`

Each file should contain properly formatted data.

Task 2

Read the `students.txt` file and display all student names.

Task 3

Read the `grades.txt` file and display all grades for the **Python** subject only.

Task 4

Ask the user to enter a `student_id`, then display:

- Student name
 - All subjects and corresponding grades
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Bonus Task (Optional)

Calculate and display the **average grade** for each student.

Notes

- Use file handling functions such as `open()`, `read()`, and `split()`
 - Handle data carefully and make sure your output is clear and readable
 - Follow clean coding practices and add comments where needed
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Skills You Will Practice

- File handling in Python
 - Data parsing
 - Problem solving
 - Writing clean and readable code
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